

label, l, k *field label*
index, i, j, n, m *indices*
termvar, x
basetype, A

terminals	$::=$ $\rightarrow$ $\rightarrow$ $\Rightarrow$ λ $\mapsto$ $\vdash$ $\emptyset$ $\times$ $<:$ $\langle$ $\rangle$		
Γ	$::=$ $\emptyset$ $\Gamma, x : T$		<i>contexts:</i> <i>empty context</i> <i>term variable binding</i>
labels	$::=$ $l_1 .. l_n$		
formula	$::=$ judgement $x : T \in \Gamma$ DISTINCT $l_1 .. l_n$ $formula_1 .. formula_n$ j INDEXES $t_1 .. t_n$ j INDEXES $T_1 .. T_n$		
C	$::=$ $\langle l = x \rangle \Rightarrow t$	bind x in t	
T, S, U	$::=$ $T \rightarrow T'$ (T) A Bool Nat Unit $T_1 + T_2$ $\langle l_i : T_i^{i \in 1..n} \rangle$ $T_1 \times T_2$ $\{ T_i^{i \in 1..n} \}$ $\{ l_i : T_i^{i \in 1..n} \}$	M	<i>types:</i> <i>type of functions</i> <i>base type</i> <i>type of booleans</i> <i>type of natural numbers</i> <i>unit type</i> <i>sum type</i> <i>type of variants</i> <i>product type</i> <i>tuple type</i> <i>type of records</i>
v	$::=$ $\lambda x : T. t$ true false 0	bind x in t	<i>values:</i> <i>abstraction value</i> <i>true value</i> <i>false value</i> <i>zero value</i>

		$\text{succ } v$ unit $\text{inl } v$ $\text{inr } v$ $\langle l = v \rangle \text{ as } T$ $\{v_1, v_2\}$ $\{v_i\}_{i \in 1..n}$ $\{l_i = v_i\}_{i \in 1..n}$	<i>successor value</i> <i>unit value</i> <i>tagged value (left)</i> <i>tagged value (right)</i> <i>tagged value</i> <i>pair value</i> <i>tuple value</i> <i>record value</i>
t	$::=$	(t) x $\lambda x : T. t$ $t_1 t_2$ $[x \mapsto t']t$ true false $\text{if } t_1 \text{ then } t_2 \text{ else } t_3$ 0 $\text{succ } t$ $\text{pred } t$ $\text{iszero } t$ unit $t_1; t_2$ $t \text{ as } T$ inert_T $\text{let } x = t \text{ in } t'$ $\text{inl } t$ $\text{inr } t$ $\text{case } t \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2$ $\langle l = t \rangle \text{ as } T$ $\text{case } t \text{ of } C_i_{i \in 1..n}$ $\{t_1, t_2\}$ $t.1$ $t.2$ $\{t_i\}_{i \in 1..n}$ $t.i$ $\{l_i = t_i\}_{i \in 1..n}$ $t.l$ $\text{fix } t$	<i>terms:</i> M <i>variable</i> $\text{bind } x \text{ in } t$ <i>abstraction</i> <i>application</i> M <i>constant true</i> <i>constant false</i> <i>conditional</i> <i>constant zero</i> <i>successor</i> <i>predecessor</i> <i>zero test</i> <i>constant unit</i> <i>sequencing</i> <i>ascription</i> <i>inert element of type T</i> $\text{bind } x \text{ in } t'$ <i>let binding</i> <i>tagging (left)</i> <i>tagging (right)</i> $\text{bind } x_1 \text{ in } t_1$ $\text{bind } x_2 \text{ in } t_2$ <i>case</i> <i>tagging</i> <i>case</i> <i>pair</i> <i>first projection</i> <i>second projection</i> <i>tuple</i> <i>projection</i> <i>record</i> <i>projection</i> <i>fixed point of t</i>
Jop	$::=$	$t \longrightarrow t'$	<i>Evaluation</i>
Jtype	$::=$	$\Gamma \vdash t : T$	<i>Typing</i>
judgement	$::=$	Jop	

	Jtype
user_syntax ::=	
	label
	<i>index</i>
	termvar
	basetype
	terminals
	Γ
	labels
	formula
	C
	T
	v
	t

$\boxed{t \longrightarrow t'}$ *Evaluation*

$$\begin{array}{c}
\frac{t_1 \longrightarrow t'_1}{t_1 t_2 \longrightarrow t'_1 t_2} \quad \text{E_APP1} \\
\frac{t_2 \longrightarrow t'_2}{v_1 t_2 \longrightarrow v_1 t'_2} \quad \text{E_APP2} \\
\frac{}{(\lambda x : T.t_{12}) v_2 \longrightarrow [x \mapsto v_2] t_{12}} \quad \text{E_APPAbs} \\
\frac{}{\text{if true then } t_2 \text{ else } t_3 \longrightarrow t_2} \quad \text{E_IFTTrue} \\
\frac{}{\text{if false then } t_2 \text{ else } t_3 \longrightarrow t_3} \quad \text{E_IFFalse} \\
\frac{t_1 \longrightarrow t'_1}{\text{if } t_1 \text{ then } t_2 \text{ else } t_3 \longrightarrow \text{if } t'_1 \text{ then } t_2 \text{ else } t_3} \quad \text{E_IF} \\
\frac{t_1 \longrightarrow t'_1}{\text{succ } t_1 \longrightarrow \text{succ } t'_1} \quad \text{E_SUCC} \\
\frac{}{\text{pred } 0 \longrightarrow 0} \quad \text{E_PREDZERO} \\
\frac{}{\text{pred } (\text{succ } v_1) \longrightarrow v_1} \quad \text{E_PREDSUCC} \\
\frac{t_1 \longrightarrow t'_1}{\text{pred } t_1 \longrightarrow \text{pred } t'_1} \quad \text{E_PRED} \\
\frac{}{\text{iszero } 0 \longrightarrow \text{true}} \quad \text{E_ISZEROZERO} \\
\frac{}{\text{iszero } (\text{succ } v_1) \longrightarrow \text{false}} \quad \text{E_ISZEROSUCC} \\
\frac{t_1 \longrightarrow t'_1}{\text{iszero } t_1 \longrightarrow \text{iszero } t'_1} \quad \text{E_ISZERO} \\
\frac{t_1 \longrightarrow t'_1}{t_1; t_2 \longrightarrow t'_1; t_2} \quad \text{E_SEQ} \\
\frac{}{\text{unit}; t_2 \longrightarrow t_2} \quad \text{E_SEQNEXT}
\end{array}$$

$$\begin{array}{c}
\frac{}{v_1 \text{ as } \top \longrightarrow v_1} \quad \text{E_ASCRIBE} \\
\\
\frac{t_1 \longrightarrow t'_1}{t_1 \text{ as } \top \longrightarrow t'_1 \text{ as } \top} \quad \text{E_ASCRIBE1} \\
\\
\frac{}{\text{let } x = v_1 \text{ in } t_2 \longrightarrow [x \mapsto v_1]t_2} \quad \text{E_LETV} \\
\\
\frac{t_1 \longrightarrow t'_1}{\text{let } x = t_1 \text{ in } t_2 \longrightarrow \text{let } x = t'_1 \text{ in } t_2} \quad \text{E_LET} \\
\\
\frac{}{\text{case } (\text{inl } v_0) \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 \longrightarrow [x_1 \mapsto v_0]t_1} \quad \text{E_CASEINL} \\
\\
\frac{}{\text{case } (\text{inr } v_0) \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 \longrightarrow [x_2 \mapsto v_0]t_2} \quad \text{E_CASEINR} \\
\\
\frac{t_0 \longrightarrow t'_0}{\text{case } t_0 \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 \longrightarrow \text{case } t'_0 \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2} \quad \text{E_CASESM} \\
\\
\frac{t_1 \longrightarrow t'_1}{\text{inl } t_1 \longrightarrow \text{inl } t'_1} \quad \text{E_INL} \\
\\
\frac{t_1 \longrightarrow t'_1}{\text{inr } t_1 \longrightarrow \text{inr } t'_1} \quad \text{E_INR} \\
\\
\frac{j \text{ INDEXES } t_i^{i \in 1..n}}{\text{case } (\langle l_j = v \rangle \text{ as } \top) \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n} \longrightarrow [x_j \mapsto v]t_j} \quad \text{E_CASEVARIANT} \\
\\
\frac{t \longrightarrow t'}{\text{case } t \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n} \longrightarrow \text{case } t' \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n}} \quad \text{E_CASE} \\
\\
\frac{t_i \longrightarrow t'_i}{\langle l_i = t_i \rangle \text{ as } \top \longrightarrow \langle l_i = t'_i \rangle \text{ as } \top} \quad \text{E_VARIANT} \\
\\
\frac{}{\{v_1, v_2\}.1 \longrightarrow v_1} \quad \text{E_PAIRBETA1} \\
\\
\frac{}{\{v_1, v_2\}.2 \longrightarrow v_2} \quad \text{E_PAIRBETA2} \\
\\
\frac{t_1 \longrightarrow t'_1}{t_1.1 \longrightarrow t'_1.1} \quad \text{E_PROJ1} \\
\\
\frac{t_1 \longrightarrow t'_1}{t_1.2 \longrightarrow t'_1.2} \quad \text{E_PROJ2} \\
\\
\frac{t_1 \longrightarrow t'_1}{\{t_1, t_2\} \longrightarrow \{t'_1, t_2\}} \quad \text{E_PAIR1} \\
\\
\frac{t_2 \longrightarrow t'_2}{\{v_1, t_2\} \longrightarrow \{v_1, t'_2\}} \quad \text{E_PAIR2} \\
\\
\frac{j \text{ INDEXES } v_i^{i \in 1..n}}{\{v_i^{i \in 1..n}\}.j \longrightarrow v_j} \quad \text{E_PROJTUPLE} \\
\\
\frac{t_1 \longrightarrow t'_1}{t_1.i \longrightarrow t'_1.i} \quad \text{E_PROJTP} \\
\\
\frac{t \longrightarrow t'}{\{v_i^{i \in 1..m}, t, t_j^{j \in 1..n}\} \longrightarrow \{v_i^{i \in 1..m}, t', t_j^{j \in 1..n}\}} \quad \text{E_TUPLE}
\end{array}$$

$$\begin{array}{c}
\frac{}{\{l_i = v_i^{i \in 1..n}\}.l_j \longrightarrow v_j} \text{E_PROJRCd} \\
\frac{t_1 \longrightarrow t'_1}{t_1.l \longrightarrow t'_1.l} \text{E_PROJ} \\
\frac{t \longrightarrow t'}{\{l_i = v_i^{i \in 1..m}, l = t, l'_j = t_j^{j \in 1..n}\} \longrightarrow \{l_i = v_i^{i \in 1..m}, l = t', l'_j = t_j^{j \in 1..n}\}} \text{E_RCd} \\
\frac{}{\text{fix}(\lambda x : T_1.t_2) \longrightarrow [x \mapsto (\text{fix}(\lambda x : T_1.t_2))]t_2} \text{E_FIXBETA} \\
\frac{t_1 \longrightarrow t'_1}{\text{fix } t_1 \longrightarrow \text{fix } t'_1} \text{E_FIX}
\end{array}$$

$\boxed{\Gamma \vdash t : T}$ *Typing*

$$\begin{array}{c}
\frac{x : T \in \Gamma}{\Gamma \vdash x : T} \text{T_VAR} \\
\frac{\Gamma, x : T_1 \vdash t_2 : T_2}{\Gamma \vdash \lambda x : T_1.t_2 : T_1 \rightarrow T_2} \text{T_ABS} \\
\frac{\Gamma \vdash t_1 : T_{11} \rightarrow T_{12} \quad \Gamma \vdash t_2 : T_{11}}{\Gamma \vdash t_1 t_2 : T_{12}} \text{T_APP} \\
\frac{}{\Gamma \vdash \text{true} : \text{Bool}} \text{T_TRUE} \\
\frac{}{\Gamma \vdash \text{false} : \text{Bool}} \text{T_FALSE} \\
\frac{\Gamma \vdash t_1 : \text{Bool} \quad \Gamma \vdash t_2 : T \quad \Gamma \vdash t_3 : T}{\Gamma \vdash \text{if } t_1 \text{ then } t_2 \text{ else } t_3 : T} \text{T_IF} \\
\frac{}{\Gamma \vdash 0 : \text{Nat}} \text{T_ZERO} \\
\frac{\Gamma \vdash t_1 : \text{Nat}}{\Gamma \vdash \text{succ } t_1 : \text{Nat}} \text{T_SUCC} \\
\frac{\Gamma \vdash t_1 : \text{Nat}}{\Gamma \vdash \text{pred } t_1 : \text{Nat}} \text{T_PRED} \\
\frac{\Gamma \vdash t_1 : \text{Nat}}{\Gamma \vdash \text{iszero } t_1 : \text{Bool}} \text{T_ISZERO} \\
\frac{}{\Gamma \vdash \text{unit} : \text{Unit}} \text{T_UNIT} \\
\frac{\Gamma \vdash t_1 : \text{Unit} \quad \Gamma \vdash t_2 : T_2}{\Gamma \vdash t_1; t_2 : T_2} \text{T_SEQ} \\
\frac{\Gamma \vdash t_1 : T}{\Gamma \vdash t_1 \text{ as } \overline{T} : \overline{T}} \text{T_ASCRIBE} \\
\frac{}{\Gamma \vdash \text{inert}_T : T} \text{T_INERT}
\end{array}$$

$$\begin{array}{c}
\frac{\Gamma \vdash t_1 : T_1 \quad \Gamma, x : T_1 \vdash t_2 : T_2}{\Gamma \vdash \text{let } x = t_1 \text{ in } t_2 : T_2} \quad \text{T_LET} \\
\\
\frac{\Gamma \vdash t_1 : T_1}{\Gamma \vdash \text{inl } t_1 : T_1 + T_2} \quad \text{T_INL} \\
\\
\frac{\Gamma \vdash t_1 : T_2}{\Gamma \vdash \text{inr } t_1 : T_1 + T_2} \quad \text{T_INR} \\
\\
\frac{\Gamma \vdash t_0 : T_1 + T_2 \quad \Gamma, x_1 : T_1 \vdash t_1 : T \quad \Gamma, x_2 : T_2 \vdash t_2 : T}{\Gamma \vdash \text{case } t_0 \text{ of inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 : T} \quad \text{T_CASESM} \\
\\
\frac{\Gamma \vdash t : T_j \quad j \text{ INDEXES } T_i^{i \in 1..n}}{\Gamma \vdash \langle l_j = t \rangle \text{ as } \langle l_i : T_i^{i \in 1..n} \rangle : \langle l_i : T_i^{i \in 1..n} \rangle} \quad \text{T_VARIANT} \\
\\
\frac{\Gamma \vdash t : \langle l_i : T_i^{i \in 1..n} \rangle \quad \Gamma, x_i : T_i \vdash t_i : T^{i \in 1..n} \quad \text{DISTINCT } l_i^{i \in 1..n}}{\Gamma \vdash \text{case } t \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n} : T} \quad \text{T_CASE} \\
\\
\frac{\Gamma \vdash t_1 : T_1 \quad \Gamma \vdash t_2 : T_2}{\Gamma \vdash \{t_1, t_2\} : T_1 \times T_2} \quad \text{T_PAIR} \\
\\
\frac{\Gamma \vdash t_1 : T_1 \times T_2}{\Gamma \vdash t_{1.1} : T_1} \quad \text{T_PROJ1} \\
\\
\frac{\Gamma \vdash t_1 : T_1 \times T_2}{\Gamma \vdash t_{1.2} : T_2} \quad \text{T_PROJ2} \\
\\
\frac{\Gamma \vdash t_i : T_i^{i \in 1..n}}{\Gamma \vdash \{t_i^{i \in 1..n}\} : \{T_i^{i \in 1..n}\}} \quad \text{T_TUPLE} \\
\\
\frac{\Gamma \vdash t_1 : \{T_i^{i \in 1..n}\} \quad j \text{ INDEXES } T_i^{i \in 1..n}}{\Gamma \vdash t_{1.j} : T_j} \quad \text{T_PROJTP} \\
\\
\frac{\Gamma \vdash t_i : T_i^{i \in 1..n} \quad \text{DISTINCT } l_i^{i \in 1..n}}{\Gamma \vdash \{l_i = t_i^{i \in 1..n}\} : \{l_i : T_i^{i \in 1..n}\}} \quad \text{T_RCD} \\
\\
\frac{\Gamma \vdash t_1 : \{l_i : T_i^{i \in 1..n}\}}{\Gamma \vdash t_{1.l_j} : T_j} \quad \text{T_PROJ} \\
\\
\frac{\Gamma \vdash t_1 : T_1 \rightarrow T_1}{\Gamma \vdash \text{fix } t_1 : T_1} \quad \text{T_FIX}
\end{array}$$

Definition rules: 69 good 0 bad
 Definition rule clauses: 130 good 0 bad